Program 1. Working with NumPy Arrays

```
import numpy as np
a = np.array([10, 20, 30, 40, 50])
b = np.array([1, 2, 3, 4, 5])
print("Array A:", a)
print("Array B:", b)
print("Shape:", a.shape)
print("Size:", a.size)
print("Dimensions:", a.ndim)
print("Data Type:", a.dtype)
print("First element:", a[0])
print("Last element:", a[-1])
print("Slicing:", a[1:4])
print("Addition:", a + b)
print("Subtraction:", a - b)
print("Multiplication:", a * b)
print("Division:", a / b)
print("Sum:", np.sum(a))
print("Mean:", np.mean(a))
print("Maximum:", np.max(a))
print("Minimum:", np.min(a))
c = np.arange(1, 10)
print("Original Array:", c)
print("Reshaped Array:")
print(c.reshape(3, 3))
```

program 2 : Working with Pandas data frames

```
 import pandas as pd
data = {
'Name': ['Arun', 'Bala', 'Chitra', 'Divya', 'Elango'],
'Age': [21, 22, 20, 21, 23],
'Department': ['M.Sc CS', 'M.Sc CS', 'M.Sc CS', 'M.Sc CS', 'M.Sc CS'],
'Marks': [85, 78, 92, 88, 75]
}
df = pd.DataFrame(data)
print("Student DataFrame:")
print(df)
print("\nNames:")
print(df['Name'])
print("\nMarks:")
print(df['Marks'])
df['Result'] = ['Pass', 'Pass', 'Pass', 'Pass', 'Pass']
print("\nAfter Adding Result Column:")
print(df)
print("\nFirst Three Rows:")
print(df.head(3))
print("\nStudents with Marks > 80:")
print(df[df['Marks'] > 80])
print("\nAverage Marks:")
print(df['Marks'].mean())
print("\nHighest Marks:")
print(df['Marks'].max())
```

```
print("\nLowest Marks:")
print(df['Marks'].min())
print("\nStudents Sorted by Marks:")
print(df.sort_values('Marks', ascending=False))
```

program 3: Basic Plots using matplotlib

```
import matplotlib.pyplot as plt

subjects = ['Python', 'Java', 'DBMS', 'Web', 'AI']

marks = [85, 78, 90, 75, 88]

plt.plot(subjects, marks, marker='o')

plt.title('Student Marks - Line Plot')

plt.xlabel('Subjects')

plt.ylabel('Marks')

plt.grid(True)

plt.show()

plt.bar(subjects, marks)

plt.title('Student Marks - Bar Chart')

plt.xlabel('Subjects')

plt.ylabel('Marks')

plt.show()

plt.scatter(subjects, marks)

plt.title('Student Marks - Scatter Plot')

plt.xlabel('Subjects')

plt.ylabel('Marks')

plt.show()

plt.pie(marks, labels=subjects, autopct='%1.1f%%')

plt.title('Student Marks - Pie Chart')

plt.show()
```

program4: Frequency Distribution, Averages, Variability

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
marks = [45, 50, 55, 60, 60, 65, 70, 70, 70, 75,
80, 80, 85, 90, 95]
data = pd.Series(marks)
frequency = data.value_counts().sort_index()
print("FREQUENCY DISTRIBUTION")
print(frequency)
mean = np.mean(marks)
median = np.median(marks)
mode = data.mode()[0]
print("\nMEASURES OF AVERAGE")
print("Mean   :", mean)
print("Median :", median)
print("Mode   :", mode)
data_range = np.max(marks) - np.min(marks)
variance = np.var(marks)
std_deviation = np.std(marks)
print("\nMEASURES OF VARIABILITY")
print("Range :", data_range)
print("Variance :", variance)
print("Standard Deviation :", std_deviation)
plt.bar(frequency.index.astype(str), frequency.values)
plt.title("Frequency Distribution of Marks")
```

```
plt.xlabel("Marks")
plt.ylabel("Frequency")
plt.grid(axis='y')
plt.show()
```